

A Human-in-the-Loop Conceptual Design Framework Jointly Driven by Large Language Models and Knowledge Graphs

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Abstract - In mechanical product development, conceptual design is a critical early stage where comprehensively eliciting user requirements and generating viable solutions are paramount. While large language models (LLMs) can surpass the innovation constraints of traditional knowledge-based design typically restricted by finite knowledge bases, their practical engineering adoption is hindered by the domain specificity, factual unreliability, and imprecise requirement interpretation. To address these challenges, this paper proposes KnowLoop, a human-in-the-loop conceptual design framework that synergistically integrates LLMs, domain knowledge graphs, and expert supervision. KnowLoop first extracts and structures requirements using slot filling based on the function-behavior-structure ontology. Next, a domain knowledge graph is constructed via chain-of-thought prompting, and its embedding representation is optimized through relational graph attention. Finally, conceptual solutions are generated guided by the knowledge graph and expert supervision. Experiments across three design tasks show that our framework outperforms state-of-the-art methods, yielding design solutions with superior responsiveness, innovativeness, feasibility, and cost efficiency.

Keywords: conceptual design; domain knowledge graphs; large language models; expert supervision.

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Abstract

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1. Introduction

In mechanical product development, conceptual design is a critical early phase that determines the innovation and performance of a product (Pahl and Beitz, 1988; Ulrich and Eppinger, 2016). In the era of Industry 4.0, driven by the demand for rapid, large-scale customization, automating and accelerating this phase has become critical (Pech, 2022). Success in conceptual design hinges on two fundamental aspects: (1) the precise modeling and mining of user requirements, which are often implicit and ambiguous; and (2) the efficient generation of viable solutions based on domain knowledge (Chen, 2010; Vuletic et al., 2018; Wang and Tseng, 2011).

Existing frameworks for requirement mining can be typically categorized into rule-based manual methods and data-driven approaches facilitated by natural language processing (NLP), as the latter has gained more attention due to its automation capabilities. For example, Wu et al. (2022) proposed a multi-source mapping method to classify and decompose heterogeneous requirements. Kwon et al. (2024) applied NLP techniques to extract structured requirements from unstructured design guidelines. Ettrich et al. (2024) developed a decision support tool to convert a binary requirement classification problem to a token classification problem, enabling the extraction of specific requirements. Necula et al. (2024) systematically reviewed the evolution of the NLP application in requirement mining and clarified key technical themes and future directions. Jin et al. (2025) proposed a framework that leveraged LLMs to simulate users and extract user requirements, effectively identifying latent needs at a lower cost than traditional user research. However, due to the limited scope of training data, data-driven methods struggle with divergent reasoning and uncovering implicit knowledge within complex, dynamic design contexts. Also, to ensure the validity and completeness of requirements, the method of transforming ambiguous user requirements into structured, domain-aligned requirement elements remains to be investigated.

Following requirement mining, the solution generation aims to transform these refined requirements into concrete design concepts. Existing methods can be broadly classified into three types: theory-driven, knowledge-driven, and data-driven. Theory-driven approaches, such as the function-based synthesis (Wood and Greer, 2001), realize structured decomposition but could be rigid and lack flexibility, while knowledge-driven methods typically leverage a knowledge base. For example, Luo et al. (2021) established a technology semantic network using patents to identify underexplored technical domains to support concept generation. However, these methods are often constrained by the boundaries of the knowledge base, limiting divergent thinking, which shifts the focus of researchers into data-driven generative models in recent years. For example, Zhu and Luo (2022a) proposed to combine the CreativeGAN with novelty detection and GAN rewriting mechanisms to guide the GAN to generate novel designs. Heyrani Nobari et al. (2021) introduced CreativeGAN to generate novel bicycle designs. Yet, despite their creativity, it could be challenging for generative models to adequately integrate critical design constraints and domain-specific logic.

Recently, LLMs have emerged as powerful tools for supporting conceptual design, valued for their strong capacities in language understanding, analogical reasoning, and creative generation. For instance, Zhu and Luo (2022b) proposed a method of fine-tuning a generative pre-trained Transformer using the USPTO patent database and they (Zhu and Luo, 2023) further utilized GPT-2 and GPT-3 for domain-specific concept synthesis. Ma et al. (2023) demonstrated that GPT-3 could generate solutions more feasible and useful than those from crowdsourcing. Similarly, Wang et al. (2023) proposed a structured reasoning approach using LLMs, and Chen et al. (2024a) incorporated LLMs into morphological analysis to expand the design space and mitigate the workload of designers. Despite this promise, since LLMs are prone to hallucinations, exhibit weak domain grounding, and offer low interpretability of reasoning, their applicability in product design is still limited in practice. These limitations raise significant concerns regarding a controllable, domain-aware solution generation method with human-in-the-loop supervision.

To overcome these limitations, this paper proposes KnowLoop, a systematic framework incorporating the creative ideation capability of LLMs, the structured reliability of domain knowledge graphs (KGs), and the critical validation of human expertise. This three-way collaboration leverages their complementary advantages in generation, knowledge retrieval, and supervision. The primary contributions of this paper are presented below:

- 1) A novel Function-Behavior-Structure (FBS)-based slot filling (SF) method is proposed to transform ambiguous design requirements into structured and complete requirement units, ensuring the domain consistency of the elicited requirements.

- 2) A KG construction method guided by chain-of-thought (CoT) prompting is proposed to improve the quality of KGs, where a relational graph attention (rGAT) network is introduced to boost knowledge representations of KGs in complex engineering design tasks.

- 3) A human-in-the-loop design workflow is established, involving continuous expert supervision (ES) over both knowledge construction and solution generation, to ensure the feasibility and validity of the final design.

The rest of this paper is organized as follows. Section 2 presents preliminary theories involved in the considered framework. Section 3 details the proposed KnowLoop methodology. Section 4 presents and analyzes the experimental results. Finally, Section 5 discusses the findings and outlines future work.

2. Preliminaries

2.1 Slot Filling (SF) Mechanism

SF (Tur and De Mori, 2011) is a fundamental task in natural language understanding (NLU), critical for extracting structured information from unstructured text, particularly within task-oriented dialogue systems. Its primary objective is to identify and categorize specific entities or attributes, referred to as “slots”, from a given natural language text.

Formally, SF is treated as a sequence labeling problem. Given an input sequence of n words, $X = [x_1, x_2, \dots, x_n]$, the objective is to assign a slot label to each word, resulting in a label sequence $Y = [y_1, y_2, \dots, y_n]$, where each y_i denotes the semantic role of the corresponding word x_i .

2.2 Chain-of-thought (CoT) Prompting

CoT reasoning (Wei et al., 2022), emulating human cognitive processes for solving complex problems, is a technique intended to explicitly guide LLMs through intermediate reasoning steps. Its core lies in using few-shot prompting to instruct the model to “think aloud” by generating a structured sequence of thoughts before providing a final answer. CoT significantly enhances LLM performance on tasks by allowing to break down complex problems, track intermediate states and reduce errors.

Formally, given an input query Q , the CoT process generates a sequence of intermediate reasoning steps iteratively, denoted by $R = [r_1, r_2, \dots, r_n]$ with r_i referring to the i -th reasoning result, and finally leads to the final answer A as expressed in (1):

$$Q \xrightarrow{\text{CoT}} r_1 \rightarrow r_2 \rightarrow \dots \rightarrow r_n \rightarrow A \quad (1)$$

3. Methodology

This paper proposes KnowLoop to achieve efficient, reliable, and innovative conceptual designs. The framework of the KnowLoop, as shown in Fig. 1, is composed of two major modules: the requirement mining module and the solution generation module. Details are provided in the following sub-sections.

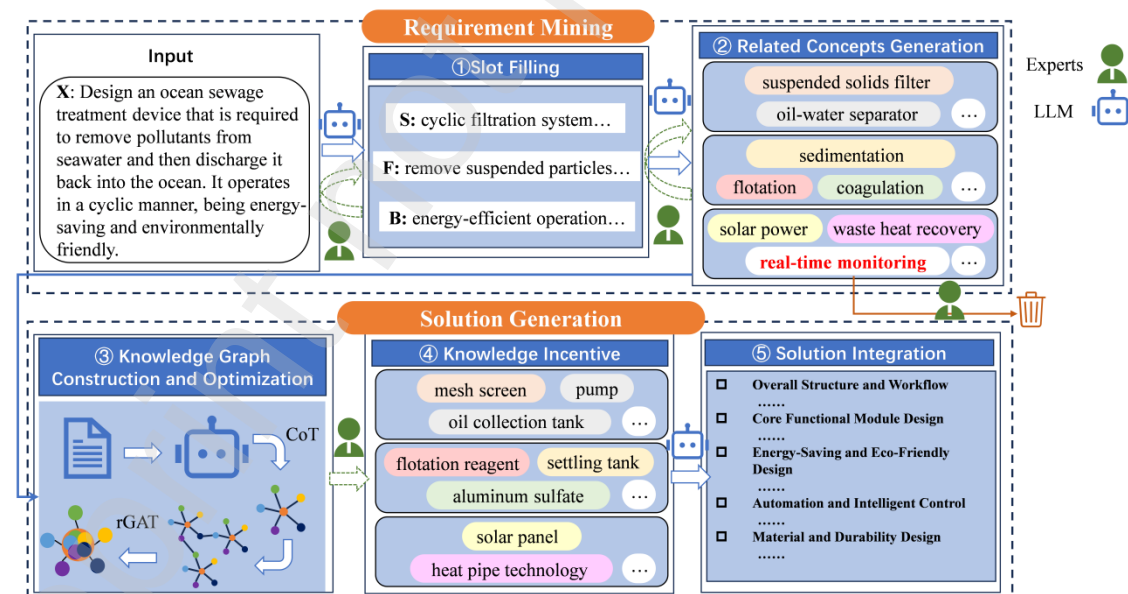


Fig. 1. The overall framework of the KnowLoop conceptual design method.

3.1 Requirement Mining

(1) SF for requirement mining

The requirement mining process is illustrated in Fig. 2. To effectively identify and structure explicit and implicit user requirements for downstream design tasks, this section introduces the FBS ontology (Gero, 1990) as design domain knowledge into the SF schema. Specifically, the *function*, *behavior*, and *structure* are designated as slots. A prompting strategy is then employed to guide LLMs in performing this SF task, i.e. to decompose the requirements into corresponding slots.

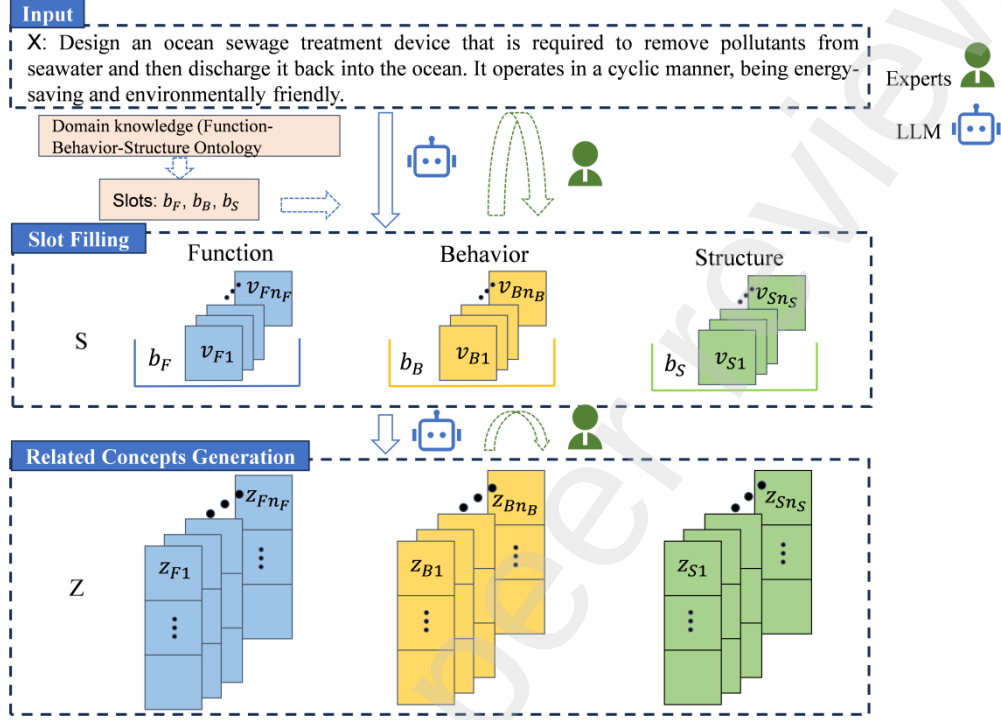


Fig. 2. SF process for user requirement mining

Formally, let X denote the input natural language requirement. We define $B = \{b_F, b_B, b_S\}$ as the predefined set of slot keys, where b_F , b_B and b_S correspond to the *function*, *behavior*, and *structure* slots, respectively. Let $p_F = \{p_{F,1}, \dots, p_{F,n_F}\}$, $p_B = \{p_{B,1}, \dots, p_{B,n_B}\}$ and $p_S = \{p_{S,1}, \dots, p_{S,n_S}\}$ denote the values extracted for each slot, and n_F , n_B , and n_S represent the number of filled values for slots b_F , b_B , and b_S , respectively. The resulting set of key-value pairs is denoted by $S = \{s_F, s_B, s_S\}$ with $s_F = (b_F, p_F)$, $s_B = (b_B, p_B)$ and $s_S = (b_S, p_S)$. Thus, the requirement mining task can be modeled as yielding S corresponding to the slot keys B from the input X , as expressed in (2) and (3):

$$X \xrightarrow{\text{LLM}} S = \{(b_F, p_F), (b_B, p_B), (b_S, p_S)\} \quad (2)$$

$$S^{(t)} = \{(b, p^{(t)}) \mid b \in B, p^{(t)} = \text{Extract}(b, X, \phi_{\text{SF}}^{(t)})\} \quad (3)$$

where $\phi_{\text{SF}}^{(t)}$ denotes the interactive prompt at the t -th iteration (provided in Appendix A), and the function $\text{Extract}(\cdot)$ refers to applying an LLM to extract values for a specific slot b from the input X guided by $\phi_{\text{SF}}^{(t)}$. The SF for requirement mining process is described in Algorithm 1 via the pseudocode.

Algorithm 1: Slot filling for requirement mining

```
1  Input:  $B$ : a predefined set of slot keys;  
2       $X$ : the original design requirement in natural language form;  
3       $\phi_{SF}$ : a prompt guiding the LLM to extract key-value pairs;  
4  Output:  $S$ : extracted key-value requirement unit pairs;  
5  Define  $B = \{\text{Function, Behavior, Structure}\}$ ;  
6  Initialize an empty key-value set:  $S = []$ ;  
7  For  $b$  in  $B$  Do  
8      If  $b$  not filled in  $B$  Then  
9           $p = \text{Extract}(b, X, \phi_{SF})$ ;  
10          $s = (b, p)$ ;  
11         add  $s$  to  $S$ ;  
12      End  
13 End
```

(2) Expert supervision (ES) for requirement mining

The ES mechanism for the SF process is expressed in (4):

$$\phi_{SF}^{(t+1)} = g(S^{(t)}, \phi_{SF}^{(t)}) \quad (4)$$

where $g(\cdot)$ denotes a correction function driven by expert rules, domain knowledge, or user preferences. This function dynamically adjusts structural, lexical, or slot constraints within the prompts to refine the extracted information. The termination condition for the iterative process is defined as (5):

$$\|S^{(t+1)} - S^{(t)}\| < \varepsilon_{SF} \quad (5)$$

where ε_{SF} represents a threshold for acceptable semantic deviation. By integrating the FBS ontology and ES, the completeness and accuracy of requirement units are guaranteed.

Subsequently, to mine implicit requirements, the LLM further generates related extended concepts z_i for each slot value $v_i \in V$ based on the filled key-value pairs S , as described in (6)-(8):

$$Z^{(t)} = \xi(S, \phi_{Exp}^{(t)}) \quad (6)$$

$$\phi_{Exp}^{(t+1)} = g(Z^{(t)}, \phi_{Exp}^{(t)}) \quad (7)$$

$$\text{if } \|Z^{(t+1)} - Z^{(t)}\| > \varepsilon_{Exp}, Z = Z^{(t)} \quad (8)$$

where $Z = \{S_F: z_F, S_B: z_B, S_S: z_S\}$, $z_F = \{z_{F,1}, \dots, z_{F,m_F}\}$, $z_B = \{z_{B,1}, \dots, z_{B,m_B}\}$, $z_S = \{z_{S,1}, \dots, z_{S,m_S}\}$ represents the whole set of implicit requirements, m_F , m_B , and m_S represent the number of elements in z_F , z_B , and z_S , respectively, and $\phi_{Exp}^{(t)}$ denotes the interactive prompt at the t -th iteration (referring to Appendix B). The iteration process terminates when the discrepancy between the generated concepts and expert expectations falls below the threshold ε_{Exp} .

3.2 Solution Generation

(1) Construction and optimization of the domain knowledge graph

Since patent data involve rich innovative ideas and technical principles in mechanical product conceptual design, we regard patent abstracts as the data source to construct the KG. To ensure the knowledge extraction accuracy and mitigate the high costs of supervised training, an automated method for constructing a mechanical domain KG (dKG) is proposed based on LLMs and CoT reasoning, as illustrated in Fig. 3, which includes three steps as detailed next.

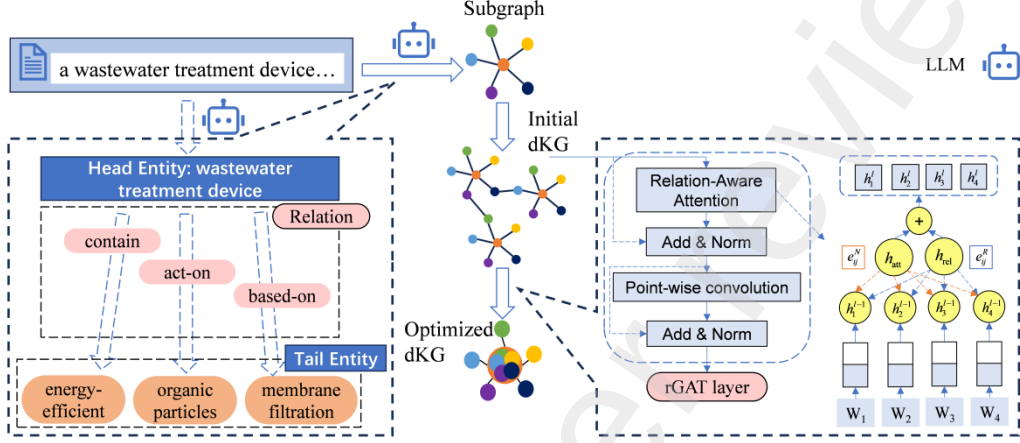


Fig. 3. The dKG construction and optimization.

(a) Subject identification

To construct the KG, we regard the unique design subject of each patent, denoted by H_i for the i -th patent abstract, as the target entity to be extracted. A patent is accordingly denoted by $d_i \in D$, $i = 1, \dots, N_D$, with N_D representing the total number of abstracts. We set H_i as the head entity of KG relations, which is expressed as (9):

$$H_i = \text{HeadIdentify}(d_i, \phi_{\text{head}}) \quad (9)$$

where ϕ_{head} is the specific prompt template for design subject entity extraction as provided in Appendix C.

(b) Relation-guided tail entity extraction

For each head entity H_i , corresponding tail entities T_{ik} are identified based on predefined relation types $r_k \in R = \{\text{contain}, \text{act_on}, \text{based_on}\}$, where R corresponds to characteristics, target objects and methods, respectively, as expressed in (10):

$$T_{ik} = \text{TailSearch}(H_i, r_k, d_i, \phi_{\text{tail}}) \quad (10)$$

where ϕ_{tail} is the prompt template guiding tail entity extraction (provided in Appendix D). In this way, a star-shaped subgraph containing one head entity and multiple tail entities is established.

Next, tail entities are recursively treated as new head entities, and their respective tail entities are sought using the same method. This recursive extraction continues until no new entities can be identified. The subgraph constructed from the i -th patent document is denoted as $G_i = (V_i, E_i)$.

(c) Entity fusion and graph merging

To consolidate knowledge from multiple subgraphs, we perform entity fusion and graph merging. For any two disconnected subgraphs $G_A = (V_A, E_A)$ and $G_B = (V_B, E_B)$, the similarity between nodes $v_i \in V_A$ and $v_j \in V_B$ is calculated using (11):

$$\text{sim}(v_i, v_j) = \frac{\vec{v}_i \cdot \vec{v}_j}{\|\vec{v}_i\| \|\vec{v}_j\|} \quad (11)$$

where $\vec{v}_i$ is the initialized word embedding vector of the node v_i . A preset similarity threshold θ_{fus} determines node equivalence. If $\text{sim}(v_i, v_j) \geq \theta_{\text{fus}}$, nodes v_i and v_j are considered equivalent and merged. Furthermore, isolated subgraphs that do not meet certain connectivity criteria are moved to ensure the overall coherence and relevance of the final dKG.

The pseudocode for domain knowledge graph construction is presented in Algorithm 2.

Algorithm 2: Domain knowledge graph construction

```

1   $G = \text{Construction}(D, \phi_{\text{head}}, \phi_{\text{tail}}, R)$ 
2  Input:  $D$ : Patent abstract set with  $N_D$  instances,  $D = \{d_1, \dots, d_{ND}\}$ ;
3       $\phi_{\text{head}}$ : prompt for head entity recognition;
4       $\phi_{\text{tail}}$ : prompt for tail entity search;
5       $R$ : predefined relation set;
6  Output:  $G$ : the expected domain knowledge graph
7  Initialize  $G = []$ ;
8  For  $i = 1$  to  $N_D$  Do
9       $H_i = \text{HeadIdentify}(d_i, \phi_{\text{head}})$ ;
10      $T_i = []$ ;
11     For each  $r$  in  $R$  Do
12          $t = \text{TailSearch}(H_i, r, d_i, \phi_{\text{tail}})$ ;
13         add  $t$  to  $T_i$ ;
14     End
15      $G_i = \text{Link}(H_i, T_i)$ ;
16     add  $G_i$  to  $G$ ;
17 End
18  $G = \text{Merge}(G, \theta_{\text{fus}})$ 
19 Input:  $G$ : all star-shaped subgraphs;
20      $\theta_{\text{fus}}$ : the node similarity threshold set by experts;
21 Output:  $G$ : updated domain knowledge graph;
22 For each  $v_i, v_j$  in  $G$  Do
23      $\text{sim} = \text{CosineSimilarity}(v_i, v_j)$ ;
24     If  $\text{sim} \geq \theta_{\text{fus}}$  Then
25         Link  $(v_i, v_j)$  in  $G$ ;
26     End
27 End

```

(2) rGAT-based embedding optimization

To capture complex dependency relations among entities and to enhance the semantic representation capability of the dKG, rGAT (Chen et al., 2020) is employed to optimize the dKG by introducing a relation-aware attention mechanism, better accommodating heterogeneous edges or relations in KGs.

Given a graph where each edge is represented as a triple, (v, r, u) with v as the target entity, u as a neighboring entity, and r indicating their relation, rGAT first projects entity and relation features into multiple semantic channels. In the k -th channel, the node features e_v^k , e_u^k and relation features e_r^k are obtained via (12)

$$e_v^k = W_e^k e_v, \quad e_u^k = W_e^k e_u, \quad e_r^k = W_r^k e_r \quad (12)$$

where W_e^k and W_r^k are parameters. The attention score of the node v with respect to its neighbor node u via relation r in the k -th channel is computed via (13)-(14)

$$\text{att}_{vru}^k = \text{LeakyReLU}(W_f^k [e_v^k \parallel e_r^k \parallel e_u^k]) \quad (13)$$

$$\alpha_{vru}^k = \frac{\exp(\text{att}_{vru}^k)}{\sum_{i \in \mu(v)} \sum_{j \in R(v)} \exp(\text{att}_{vji}^k)} \quad (14)$$

where $\text{LeakyReLU}(W_f^k[\cdot])$ is a non-linear layer, $\mu(v)$ and $R(v)$ are the set of neighboring nodes and the set of relations connected to node v in the graph, respectively. Next, the aggregated representations of the node v at layer l in the k -th channel is computed via (15) under the relation guidance.

$$e_v^{(l),k} = \sigma \left(\sum_{u \in \mu(v)} \sum_{i \in R(v)} \alpha_{vui}^k \cdot (e_u^k * e_i^k) \right) \quad (15)$$

where σ is a non-linearity and $*$ denotes the Hadamard multiplication. Finally, all channel-specific representations are concatenated to obtain the final embedding at layer l , as described in (16).

$$e_v^{(l)} = \parallel_{k=1}^K e_v^{(l),k} \quad (16)$$

By define the above process as the function $\text{rGAT}(\cdot)$, given the initial embedding $e_v^{(0)}$, $e_u^{(0)}$, $e_r^{(0)}$, the output at the l -th layer is computed as (17).

$$e_v^{(l)} = \text{rGAT}(e_v^{(l-1)}), l = 1, 2, \dots, L \quad (17)$$

where L is the number of layers in the rGAT model.

To train the rGAT model parameters, the embeddings of entities and relations are first initialized using a pre-trained model. Furthermore, a contrastive learning-based loss function is designed to jointly optimize positive and negative sample losses for robust embedding learning. The positive and negative sample losses are formulated in (18) and (19), respectively.

$$L_{\text{pos}} = \frac{1}{|\mathbf{P}|} \sum_{(i,j) \in \mathbf{P}} (1 - \text{sim}(e_i, e_j)) \quad (18)$$

$$L_{\text{neg}} = \text{ReLU}(\frac{1}{|\mathbf{N}|} \sum_{(i,j) \in \mathbf{N}} \text{sim}(e_i, e_j)) \quad (19)$$

where P denotes the set of node pairs connected by existing edges, and N denotes the set of disconnected node pairs, and $\text{ReLU}(\cdot)$ is a non-linear activation function ensuring non-negativity. The total contrastive loss is given by (20).

$$\mathcal{L}_{\text{contrastive}} = \mathcal{L}_{\text{pos}} + \mathcal{L}_{\text{neg}} \quad (20)$$

(3) Knowledge incentive

To mitigate factual hallucinations of LLMs, the constructed dKG is utilized to retrieve relevant domain-specific design knowledge related to the generated concepts $\{z_i\}$. Specifically, for each z_i , given a pre-set similarity threshold θ_{inc} , nodes satisfying the condition (21) are selected to form a candidate set Q_i :

$$Q_i = \{v_{ij} \mid \text{sim}(z_i, v_{ij}) \geq \theta_{\text{inc}}\}, j = 1, 2, \dots \quad (21)$$

where v_{ij} denotes the j -th knowledge node associated with the i -th LLM-generated concept. Experts further review Q and select the most appropriate knowledge entries, forming the final knowledge incentive set Q^* , as shown in (22).

$$Q^* = \{v_{ij}^*\} \quad (22)$$

In this way, experts can dynamically adjust the global threshold θ_{inc} to strike a balance between the retrieval breadth and knowledge relevance.

(4) LLM-based conceptual solution generation

Based on the term set Z and the knowledge set Q^* , the conceptual design scheme W is generated by using prompt ϕ_{SG} (provided in Appendix E). Formally, the generation process can be represented as (23):

$$W = \text{GenerateDesign}(Z, Q^*, \phi_{\text{SG}}) \quad (23)$$

4. Experimental Studies

4.1 Data Description and Hyperparameter Settings

In this section, experiments are conducted to validate the proposed KnowLoop method. Specifically, mechanical conceptual design tasks on three types of water treatment equipment are considered: *a domestic sewage treatment device*, *a marine sewage treatment device*, and *a medical sewage treatment device*.

To construct the dKG, over 2,000 patent abstracts from the United States Patent and Trademark Office (USPTO) in 2020 are utilized as the corpus. GPT-4o (Open AI) is employed to construct the dKG, and Deepseek-R1 (Guo et al., 2025) is used to generate the conceptual design solutions. The Word2Vec model (Mikolov et al., 2013) is utilized to initialize the word embeddings of the nodes and edges in the dKG. Hyperparameter settings of the rGAT are presented in Table 1.

Table 1. Hyperparameter settings

Hyperparameters	Values
Training epochs	500
dropout	0.3
alpha	0.5
the number of attention heads	4
hidden layer dimension	16
learning rate	10^{-5}
similarity threshold of knowledge	0.92
retrieval	
weight decay	0.01

4.2 Experiment settings

To demonstrate the effectiveness of the overall proposed method and each contribution separately, we compare the KnowLoop with a state-of-the-art (SOTA) model, FBStd (Chen et al., 2024b), and also conduct ablation studies. The FBStd was proposed to guide LLMs to generate conceptual designs through a three-stage prompt engineering based on FBS. To perform ablation studies, three experiments are conducted as follows:

B-1 (KnowLoop-w/o-SF): To verify the effectiveness of the SF mechanism in improving the completeness of requirement mining, this method replaces the SF-based requirement identification with the prompts provided by FBStd in the considered framework.

B-2 (KnowLoop-w/o-rGAT): To verify the effectiveness of the rGAT, this method applies directly the Word2Vec model for the knowledge representation in the proposed framework.

B-3 (KnowLoop-w/o-ES): To verify the significance of ES, this method removes the ES mechanism in the entire framework.

4.3 Evaluation Metrics

Ten mechanical engineering-related experts participated in evaluating the design solution quality from four dimensions: Responsiveness, Innovativeness, Feasibility, and Cost efficiency. Particularly:

(1) Responsiveness: the degree of matching between the generated solution and the explicit requirement provided by the user.

(2) Innovativeness: the innovation degree of the generated solution compared to conventional solutions.

(3) Feasibility: the extent to which the generated solution conforms to basic engineering principles and practical implementation.

(4) Cost efficiency: the cost competitiveness of the generated solution in actual production.

We adopt the structured evaluation method of the Pugh Decision Matrix (Pugh, 1993). Taking the preset benchmark solution as a reference, the final score for each solution is determined by calculating the difference between the average expert score and the benchmark score.

4.4 Experimental Results

(1) Construction and Optimization Results of dKG

Based on the USPTO patent data, subgraphs are first generated, and a small number of isolated subgraphs are deleted after fusion. Finally, a dKG with 1,106 entities, 1,162 edges and 100% connectivity is obtained, as illustrated in Fig. 4.



Fig. 4. The illustration of the water treatment equipment dKG

In the dKG knowledge embedding optimization process, we adopted a 5-fold cross-validation strategy to systematically evaluate the model performance. The loss curves are shown in Fig. 5.

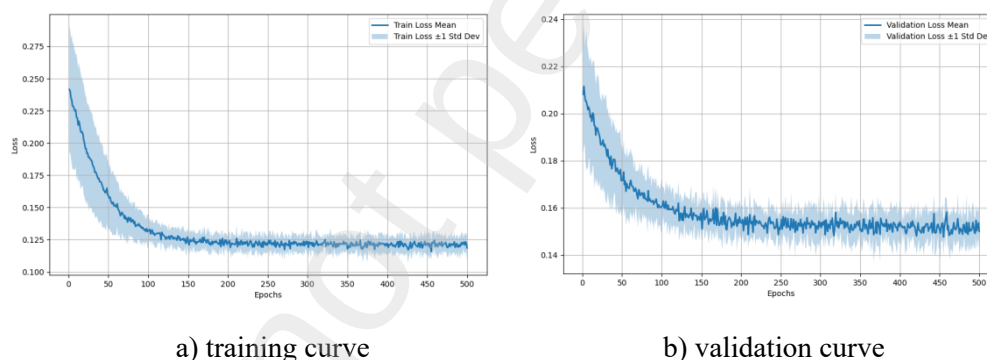


Fig. 5. The loss curves during rGAT optimization on the (a) training set and (b) validation set.

It is first observed that both the training and validation loss curves exhibit a consistent downward trend, indicating effective convergence of the rGAT model. Specifically, the training loss decreases rapidly within the initial 100 epochs and gradually stabilizes around 0.125, while the validation loss also demonstrates a sharp decline in the early stage and ultimately stabilizes at approximately 0.15 after 300 epochs. The narrow fluctuation bands (± 1 standard deviation) indicate satisfactory generalization capability and robustness of the rGAT.

(2) Conceptual design solutions

For simplicity, this section presents a part of the KnowLoop-generated and FBStd-generated design solution results using a *domestic sewage treatment device* as an example in Table 2. Complete results for all three considered devices are provided in the Appendix F.

Table 2. Conceptual design solutions for a domestic sewage treatment device

Methods	Conceptual design solutions
FBStd	• Physical Filtration: Removes plastic, solid particles, and other large pollutants. • Energy Recovery: Anaerobic digestion produces biogas for household cooking. • Physical Layer: ◦ Self-cleaning Grating: Inclined screen automatically intercepts plastics, with periodic backwashing... • Biological Layer: ◦ Moving Bed Biofilm Reactor (MBBR)... ...
KnowLoop	Filtration Unit: Multi-Stage Physical Separation <i>Function:</i> Remove plastics, grit, and suspended solids. <i>Design:</i> • Three-Stage Filtration Process: ◦ Primary Coarse Filtration: Stainless steel... ◦ Secondary Hydrocyclone Separation... ◦ Tertiary Membrane Filtration: Self-cleaning ultrafiltration membrane... • Intelligent Backwash System: ◦ Uses PLC-controlled pulse backwash technology... ◦ Pressure Monitoring: Corrosion-resistant pressure sensors... Biological Reaction Unit: Hybrid Biodegradation <i>Function:</i> Decompose organic matter and remove nitrogen/phosphorus. <i>Design:</i> • Dual-Mode Reactor: ◦ Aerobic Zone... ◦ Anaerobic Zone... Energy Recovery Unit: Thermal and Electrical Recycling <i>Function:</i> Reduce overall system energy consumption. <i>Design:</i> • Waste Heat Recovery: ◦ Plate Heat Exchanger: ... to preheat influent ... • Energy Self-Sufficiency: ◦ Microbial Fuel Cells (MFCs): ... to power sensors... ...

(3) Evaluation Results of the Solutions

1) Comprehensive Analysis of Experimental Results

Table 3 presents the scores of each solution. Compared to FBStd, the proposed KnowLoop achieved significant improvements across all four evaluation dimensions, consistently outperforming FBStd, which indicates that the proposed framework can effectively enhance the quality of conceptual design solutions. Table 4 demonstrates the average scores of each solution in these 4 dimensions.

Table 3. Scores of each solution for all methods

Products	Evaluation Dimensions	FBStd	KnowLoop	KnowLoop -w/o-SF	KnowLoop-w/o-rGAT	KnowLoop -w/o-ES
Domestic sewage treatment device	Responsiveness	82.3	86.0	83.6	85.4	86.0
	Innovativeness	79.3	82.4	78.6	82.3	82.2
	Feasibility	87.6	88.7	86.0	86.7	88.1
	Cost efficiency	81.1	82.4	78.8	79.7	80.4
Marine sewage treatment device	Responsiveness	87.1	90.8	87.4	90.4	90.5
	Innovativeness	81.9	83.3	80.4	82.9	83.3
	Feasibility	85.7	87.4	81.3	84.4	87.1
	Cost efficiency	80.1	81.3	78.0	77.9	81.3
Medical sewage treatment device	Responsiveness	85.7	89.5	85.6	88.5	82.9
	Innovativeness	81.1	84.2	77.4	82.0	84.2
	Feasibility	88.9	92.6	87.2	87.8	92.1
	Cost efficiency	83.6	83.7	82.0	83.4	83.4

Table 4. Average scores of each solution in four considered dimensions

	FBStd	KnowLoop	KnowLoop-w/o-SF	KnowLoop-w/o-rGAT	KnowLoop-w/o-ES
Responsiveness	85.0	88.8	85.5	88.1	86.5
Innovativeness	80.8	82.6	78.8	82.4	83.2
Feasibility	87.4	89.6	84.8	86.3	89.1
Cost efficiency	81.6	82.5	79.6	80.3	81.7

To statistically verify the performance differences among all considered methods, we employed the Friedman test to perform a non-parametric significance analysis on expert scores across the three tasks. The test statistic is calculated as (24)

$$\chi_F^2 = \frac{12n}{k(k+1)} \sum_{j=1}^k \left(\bar{R}_j - \frac{k+1}{2} \right)^2 \quad (24)$$

where n denotes the number of experts (10 in this study), k is the number of tasks (i.e., 3 product types $\times$ 4 evaluation dimensions), and $\bar{R}_j$ is the average rank of the j -th method. The p -value heatmap of the Friedman test (Friedman, 1940) results across each evaluation dimension and product type is shown in Fig. 6, where values $p < 0.05$ suggest statistically significant differences.

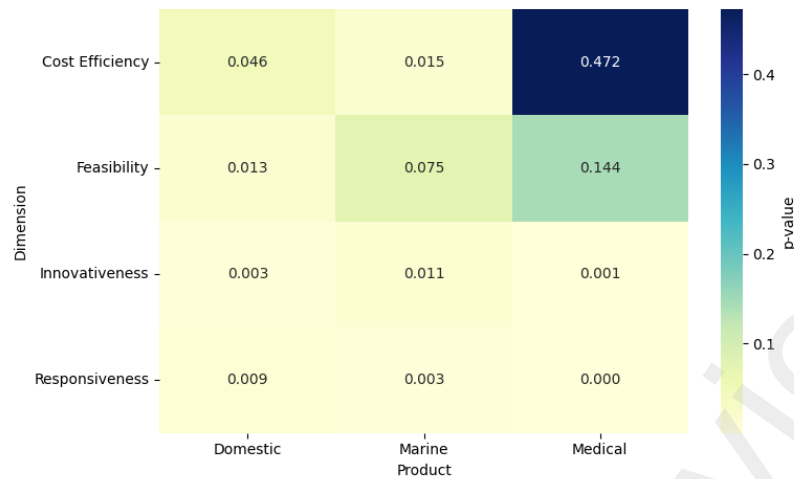


Fig. 6. Friedman test results of the improvements of the KnowLoop compared to FBStd.

The result in Fig. 6 indicates that the performance differences among the two methods are statistically significant in the majority of dimension-product combinations. Specifically, dimensions such as the innovativeness and responsiveness exhibit consistently low p -values across all three product categories, suggesting that the proposed method leads to notably more creative and accurate design responses. Additionally, statistically significant differences are also observed in the cost efficiency and feasibility dimensions for a part of products. However, higher p -values are observed in the medical product on the dimensions of cost efficiency (0.472) and feasibility (0.144). This may be attributed to the strict regulatory and functional constraints of medical product design, which often lead to similar solutions between different methods. Moreover, in highly standardized domains like healthcare, baseline methods may already capture sufficient feasibility and cost efficiency, reducing the observable margin for improvements.

As demonstrated by the results of the ablation studies (Tables 3 and 4), the proposed KnowLoop significantly outperforms KnowLoop-w/o-SF across all four evaluation dimensions, with particularly notable gains in the innovativeness. Specifically, KnowLoop achieves an average increase of 4.8% in innovativeness scores compared to its SF-ablated counterpart, with the improvement reaching 8.8% for medical sewage treatment, 3.6% for marine, and 4.8% for domestic applications. KnowLoop also yields improvements of 1.8%-2.8% in the feasibility and cost efficiency dimensions. These consistent enhancements highlight the critical role of the SF mechanism in accurately mining user requirements.

Comparatively, KnowLoop-w/o-rGAT demonstrated relatively strong performance. However, the KnowLoop still yielded notable improvements in feasibility, with an average increase of 2.6% across three products. These enhancements validate the contribution of the rGAT-based knowledge graph optimization, which facilitates more accurate propagation of semantic relations and thereby enhances the factual consistency and practicability of the generated solutions.

While KnowLoop-w/o-ES preserved competitive results in most dimensions, achieving the highest responsiveness score (90.5) for marine applications and

maintaining robust innovativeness and feasibility, it slightly lagged behind in the cost efficiency. Comparatively, the KnowLoop outperformed the ES-ablated variant by 2.5% in domestic, matching performance in marine, and showing a 0.4% gain in medical designs. Results indicate that the ES mechanism plays a subtle yet essential role in refining structural trade-offs, ensuring more cost-effective and context-appropriate design decisions, especially in resource-constrained settings.

2) Effects of Key Mechanisms on Design Performance

a. Impact of the FBS Ontology-Based SF Mechanism on Requirement Mining

Table 5 presents two examples demonstrating that KnowLoop-generated solution aligns among function, behavior, and structure. In contrast, although the FBStd-generated solution also covers elements of function, behavior, and structure, it fails to jointly model them, leading to insufficient and unaligning design description in the overall design.

Table 5. Two examples from KnowLoop-generated and FBStd-generated solutions

Instances	KnowLoop	FBStd
1	{F: remove plastics, grit, and suspended solids; B: three-stage physical isolation; S: stainless steel, hydrocyclone separation and ultra-filtration membranes}	{F: Physical Filtration; B: Removes plastic, solid particles, and other large pollutants; S: ?}
2	{F: decompose organic matter and remove nitrogen/phosphorus; B: biological reaction; S: dual-zone reactor}	{F: ?; B: ?; S: biological layer}

b. Role of the Domain Knowledge Graph in Knowledge Reuse:

Examples in Table 6 illustrate that in the KnowLoop-generated solution, key design components such as the integration of *aerobic and anaerobic zones*, a three-stage filtration chain (*coarse filtration, hydrocyclone separation, and ultrafiltration membrane*), and dual-mode energy recovery (*waste heat exchangers and MFCs*) are structurally organized based on semantic linkages encoded in the dKG, enabling KnowLoop to reconstruct the solution into highly coupled, functionally interdependent modules that enhance system-level coherence and efficiency.

In contrast, the FBStd-generated design, lacking knowledge grounding, adopts loosely connected modules such as *MBBR* and *inclined screens*, without capturing cross-module functional logic. As a result, FBStd fails to ensure semantic integration and produces fragmented outputs. This comparison highlights how the dKG-guided reasoning in KnowLoop supports the reuse and recombination of domain knowledge into coherent, knowledge-driven system designs, surpassing the conventional process-oriented generation of FBStd.

Table 6. The comparison in knowledge use between FBStd and KnowLoop

Methods	Content	Knowledge resource
KnowLoop	{aerobic zone, anaerobic zone} {coarse filtration, hydrocyclone separation, self-cleaning ultrafiltration membrane} {waste heat recovery, plate heat exchangers, MFCs}	dKG
FBStd	{MBBR, ?} {Inclined screen, ?}	LLM generation

c. The Role of ES

To demonstrate the effectiveness of ES in the KnowLoop framework, we compare it against the FBStd method across two key stages: extended concepts generation and knowledge incentive (Table 7). In the extended concepts generation stage, KnowLoop introduced *a PLC-controlled pulse backwashing system* and *pressure sensors*, whereas FBStd only proposed *periodic backwashing*. This advancement in KnowLoop significantly reduced manual maintenance and membrane replacement frequency, thereby lowering operational costs. In the knowledge incentive stage, KnowLoop incorporated *energy powering sensors* and *a preheating mechanism for influent water*, while FBStd suggested using waste heat merely *for cooking*. The enhanced solution in KnowLoop offered higher energy efficiency, indicating a deeper integration of energy recovery strategies driven by domain expertise. These examples highlight how ES constrains the solution space with domain specific heuristics and facilitates the generation of more cost-effective, robust, and sustainable design alternatives by guiding LLMs toward semantically consistent and functionally optimized solutions.

Table 7. The contrast in ES between FBStd and KnowLoop

Supervision Stages	KnowLoop	FBStd
Extended concepts generation	PLC-controlled pulse backwashing system; pressure sensors	periodic backwashing
Knowledge incentive	energy powering sensors; preheating influent water	for cooking

5. Conclusion

In this paper, we proposed a human-in-the-loop conceptual design framework that integrated LLMs and domain knowledge to address the limitations in requirement mining and domain knowledge utilization. Specifically, requirement mining was achieved through a SF mechanism grounded in the FBS ontology, enabling structured extraction of user needs. The dKG was constructed via CoT prompting to enhance the interpretability and reasoning over domain knowledge, further refined via the rGAT to optimize the embedding representation. Guided by the structured KG and ES, the framework then facilitated the conceptual solution generation that is both knowledge-

grounded and adaptable to expert feedback, ensuring a robust, iterative co-design process.

Experiments were conducted using the water treatment equipment design as an example. The results showed that, compared with the baseline method, our approach significantly improved the responsiveness, innovativeness, feasibility, and cost efficiency.

In future work, it remains to explore dynamic and continuously learning knowledge graphs that can evolve with emerging patents and technical documents. Moreover, the framework can be extended to cross-domain conceptual design tasks such as biomedical devices or autonomous systems to test its scalability and stimulate broader innovation.

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Appendix

A. The top three prompts of SF process.

t	Prompts
0	“Please construct the three slots of Function, Behavior, and Structure according to the FBS ontology theory, and decompose the following user input requirements to fill them in. Then, output the results as key-value pairs. Input: $[X]$ ”
1	“You did a great job, but please modify the values inside the slot $[]$ based on the content within the input content $[]$.”
2	“It's already close to perfect, but please consider whether the content $[]$ in the slot $[]$ meets the requirements. If it doesn't, please revise or remove it accordingly.”

B. The top three prompts of related concepts generation process.

t	Prompts
0	“Please generate some related content based on the following key-value pairs, where the “key” indicates the domain of the content, and the “value” represents the central focus around which the content should be generated.”
1	“Very good. Could you please add a few more items to the content inside $[]$?”
2	“Please consider whether the content $[]$ within $[]$ is reasonable. If not, please make the necessary modifications.”

C. Prompt of head entity extraction

“This is a patent abstract: $[d_i]$, which contains a unique design subject of the patent. Please identify that subject.”

D. Prompt of tail entities extraction

“Please locate, within the patent text, the tail entities related to the given head entity through the relations “contain”, “act_on”, and “based_on”. If necessary, you may summarize them yourself. Here, “contain” refers to the features or components included in the head entity, “act_on” refers to the objects or targets the head entity acts upon, and “based_on” indicates the principles or methods on which the development of the head entity is based.”

E. Prompt of generating conceptual design solutions

“Please generate conceptual design solutions based on the design problem [X]. The following are the related concepts and their corresponding knowledge: [Z, Q*].”

F. Complete results for all three considered design tasks

(1). Conceptual Design Solutions for a Domestic Sewage Treatment Device

Methods	Conceptual design solutions
FBStd	• Physical Filtration: Removes plastic, solid particles, and other large pollutants. • Biological Treatment: Decomposes organic waste. • Chemical/Advanced Oxidation Treatment: Degrades inorganic pollutant and harmful bacteria. • Disinfection: Kills pathogens using ultraviolet (UV) light or ozone. • Low Energy Consumption: Minimizes reliance on electricity by utilizing natural processes. • Eco-friendly Materials: Avoids the use of highly polluting chemicals; utilizes recyclable or biodegradable materials. • Intelligent Monitoring: Uses sensors to monitor water quality parameters (COD, BOD, pH, etc.) to ensure compliance with discharge or reuse standards. • Compact Structure: Suitable for installation in courtyards or basements, with a footprint of $\leq 2 \text{ m}^2$. • Low Maintenance: Includes automated sludge removal and easy-to-replace filter media. • Energy Recovery: Anaerobic digestion produces biogas for household cooking, or solar energy powers aeration pumps. • Multi-stage Treatment Process: ◦ Primary Stage: Grating + Grit chamber (removes plastics/gravel) ◦ Secondary Stage: Anaerobic reactor (decomposes organics) + Aerobic biofilm (MBBR technology) ◦ Tertiary Stage: Constructed wetland (plants absorb nitrogen/phosphorus) or activated carbon adsorption ◦ Quaternary Stage: UV disinfection unit • Physical Layer: ◦ Self-cleaning Grating: Inclined screen automatically intercepts plastics, with periodic backwashing ◦ Swirl Grit Chamber: Uses centrifugal force to separate sand and gravel, preventing downstream clogging • Biological Layer: ◦ Anaerobic Baffled Reactor (ABR): Compartmentalized chambers for organic matter degradation; biogas stored in small gas bags ◦ Moving Bed Biofilm Reactor (MBBR): Suspended carriers increase surface area for microbial attachment, efficiently reducing COD/BOD • Chemical/Ecological Layer: ◦ Vertical-flow Constructed Wetland: Plants like reeds and cattails absorb heavy metals and nutrients

	○ Photocatalytic Oxidation Unit: TiO₂-coated surfaces and LED UV lights decompose trace pharmaceutical residues • Intelligent System: ○ Arduino Control Panel: Monitors flow rate and water quality; triggers alarms and switches to backup treatment paths during anomalies ○ Solar Power Supply: Prioritizes photovoltaic panels to power pumps and UV lamps; includes battery storage
KnowLoop	System Architecture and Module Design The system is divided into five core modules, featuring a compact design suitable for household applications: • Filtration Unit (Physical Separation) • Biological Reaction Unit (Aerobic/Anaerobic Degradation) • Chemical Reaction Unit (Pollutant Neutralization) • Energy Recovery Unit (Waste Heat and Energy Reuse) • Emission Unit (Safe Discharge and Monitoring) Filtration Unit: Multi-Stage Physical Separation <i>Function:</i> Remove plastics, grit, and suspended solids. <i>Design:</i> • Three-Stage Filtration Process: ○ Primary Coarse Filtration: Stainless steel mesh screen (5 mm aperture) intercepts large particles such as plastics and fibers. ○ Secondary Hydrocyclone Separation: Centrifugal filter (Hydrocyclone) separates microplastics (<1 mm) and grit. ○ Tertiary Membrane Filtration: Self-cleaning ultrafiltration membrane (PVDF material) captures fine suspended solids (<0.1 µm). • Intelligent Backwash System: ○ Uses PLC-controlled pulse backwash technology, powered by a submersible pump and solenoid valve, for periodic membrane cleaning to prevent clogging. ○ Pressure Monitoring: Corrosion-resistant pressure sensors monitor membrane flux in real time and trigger automatic cleaning. ○ Advantages: Plastic removal rate >99%, suspended solids (TSS) <10 mg/L. ○ Low energy consumption: Uses a small electric pump (single-phase motor, power ≤100 W). Biological Reaction Unit: Hybrid Biodegradation <i>Function:</i> Decompose organic matter and remove nitrogen/phosphorus. <i>Design:</i> • Dual-Mode Reactor: ○ Aerobic Zone: ▪ Stirred Tank Reactor (FRP material): Equipped with a low-speed propeller (50 W) and microporous aeration discs for efficient mixing and oxidation of activated sludge. ▪ Biofilm Carrier: Filled with porous ceramic balls loaded with nitrifying bacteria (Nitrosomonas) and denitrifying bacteria (Pseudomonas). ○ Anaerobic Zone: ▪ Packed Bed Reactor: Contains activated carbon particles and Anammox bacteria for nitrogen removal and degradation of refractory organics. ▪ Temperature Control: Maintains 35 – 40 °C (optimal for anaerobic bacteria) through waste heat recovery. ▪ Key Parameters: COD removal rate >90%, BOD <5 mg/L, total nitrogen (TN) <15 mg/L. ○ Energy self-sufficiency: Recovers electricity from biological reactions via modular microbial fuel cells (MFCs) (power output 5–10 W/m³). Chemical Reaction Unit: Synergistic Oxidation and Precipitation <i>Function:</i> Neutralize heavy metals, inorganic salts, and residual toxins. <i>Design:</i> • Catalytic Oxidation System: ○ TiO₂ Photocatalytic Reactor: Decomposes pharmaceutical residues and organic toxins under UV-LED (365 nm wavelength) irradiation.

- Ozone Synergistic Oxidation: Generates ozone (0.5–1 mg/L) via water electrolysis to enhance degradation of persistent pollutants (e.g., PFAS).
- Chemical Precipitation:
 - Automatic Dosing System: Adds polyaluminum chloride (PAC) and polyacrylamide (PAM) to precipitate phosphates and heavy metals.
 - Dynamic pH Adjustment: Uses pH sensors and peristaltic pumps to automatically add sulfuric acid (H₂SO₄) or sodium hydroxide (NaOH), maintaining pH 6.5–7.5.
 - Advantages:
 - Heavy metal removal rate (e.g., Cr⁶⁺, Pb²⁺) >95%, phosphate (PO₄³⁻) <0.5 mg/L.
 - Chemical consumption reduced by 30% (optimized dosing frequency via AI algorithms).

Energy Recovery Unit: Thermal and Electrical Recycling

Function: Reduce overall system energy consumption.

Design:

- Waste Heat Recovery:
 - Plate Heat Exchanger: Recovers waste heat (40 – 50 °C) from the biological reactor to preheat influent or supply domestic hot water.
 - Solar Assistance: Integrates solar thermal tubes to enhance heat recovery efficiency (overall energy savings ≥20%).
- Energy Self-Sufficiency:
 - Microbial Fuel Cells (MFCs): Generate electricity from electrons produced in the anaerobic zone to power sensors and control systems.
 - Low-Power Design: Uses LED lighting and variable-frequency drive pumps, with total energy consumption ≤0.3 kWh/m³.

Emission Unit: Safe Discharge and Intelligent Monitoring

Function: Ensure effluent compliance and monitor discharge quality.

Design:

- Multi-Stage Disinfection:
 - UV Disinfection: UV-C lamps (254 nm wavelength) eliminate 99.99% of bacteria and viruses.
 - Emergency Chlorination: Automatically doses sodium hypochlorite (NaClO) to handle water quality fluctuations (residual chlorine <0.2 mg/L).
- Emission Monitoring:
 - Online Water Quality Analyzer: Real-time monitoring of COD, ammonia nitrogen, pH, etc., with data uploaded to a mobile app.
 - Gas Purification: Removes odors via activated carbon adsorption towers, with exhaust discharged through a low-emission stack.

(2). Conceptual Design Solutions for a Marine Sewage Treatment Device

Methods	Conceptual design solutions
FBStd	• Pollutant Removal: ○ Physical Filtration: Remove microplastics, sediments, and large debris. ○ Biological Treatment: Degrade organic waste (e.g., sewage, algae). ○ Chemical Treatment: Break down inorganic pollutants (e.g., heavy metals, oil spills). ○ Disinfection: Eliminate harmful bacteria (e.g., E. coli, Vibrio). • Energy Efficiency: ○ Utilize renewable energy (solar, wave, or tidal power). ○ Minimize electricity consumption (e.g., passive flow systems). • Eco-Friendly Operation: ○ No chemical additives that harm marine life. ○ Minimal sludge production (or biodegradable sludge). • Cyclic Operation: ○ Continuous intake, treatment, and discharge without manual intervention. ○ Self-cleaning mechanisms to prevent biofouling. • Modular & Scalable:

-
- Can be deployed as floating units near coastal cities or attached to ships.
 - Adjustable capacity (e.g., 1,000–10,000 L/day).
 - Low-Maintenance:
 - Automated backwashing for filters.
 - Corrosion-resistant materials (e.g., marine-grade stainless steel, HDPE).
 - Hybrid Energy Supply:
 - Solar panels for daytime power.
 - Wave energy converters (WECs) for continuous operation.
 - Closed-Loop System:
 - Treated water meets WHO/EPA standards before discharge.
 - Optional water reuse (e.g., for aquaculture or desalination).
-

KnowLoop Structural Design

- Corrosion-Resistant Main Structure: A titanium alloy shell (lightweight and resistant to seawater corrosion) with an internal hydrophobic anti-biofouling coating (to prevent marine organism attachment).
- Cyclic Filtration System: A four-stage series processing module:
 - Oil-Water Separator (centrifugal) → Suspended Solids Filter (mesh screen + sand filter) → Fixed-Bed Bioreactor → Spiral Wound Membrane Filtration System.
 - Auxiliary Units: Acid-base neutralization tank, concentrated waste liquid storage tank, and variable frequency drive pump system.

Core Functional Module Design

- Oil-Water Separation and Suspended Solids Removal
 - Centrifugal Oil-Water Separator: Separates oil and seawater via centrifugal force, with oil collected in an oil collection tank (remotely controlled for recovery).
 - Suspended Solids Filtration:
 - Primary Mesh Screen (50µm pore size) intercepts large particles;
 - Secondary Sand Filter (quartz sand + activated carbon) adsorbs fine suspended particles;
 - Chemical Coagulation Enhancement: Addition of polyaluminum chloride (PAC) to improve flocculation efficiency.
- Organic Pollutant Degradation
 - Fixed-Bed Bioreactor:
 - Filled with high-surface-area bio-carriers (e.g., polyethylene porous carriers) loaded with salt-tolerant microbial communities for organic matter degradation;
 - Electrochemical Oxidation Assistance: Anode (titanium-based lead dioxide) + cathode (graphite), generating hydroxyl radicals ($\cdot\text{OH}$) to enhance the decomposition of refractory organic pollutants (e.g., petroleum hydrocarbons).
 - Heavy Metal Separation and pH Adjustment
 - Spiral Wound Membrane Filtration System: Uses 0.1µm ceramic membranes to intercept heavy metal ions (e.g., lead, mercury), with concentrated waste stored in a waste liquid tank;
 - pH Adjustment Module: Online pH Sensor monitors in real time, automatically dosing weak alkaline solution (e.g., sodium bicarbonate) to neutralize acidic sewage.

Energy-Saving and Eco-Friendly Design

- Energy Optimization
 - Variable Frequency Drive Technology: Pumps and centrifuges use inverters to adjust speed based on real-time flow demand, reducing energy consumption;
 - Waste Heat Recovery: Heat generated by the bioreactor is used to preheat incoming water via a heat exchanger, reducing electric heating requirements.
 - Eco-Friendly Discharge
 - Nearshore Intermittent Discharge: Intelligent discharge timing based on tidal patterns to minimize ecological impact on local waters;
 - Low-Noise Design: Pumps equipped with vibration-damping bases and pipes lined with sound-absorbing materials, ensuring noise levels $\leq 55\text{dB}$;
-

- Water Quality Monitoring: Outlet equipped with multi-parameter sensors (pH, turbidity, COD, heavy metals), with automatic recycling to the treatment module if standards are exceeded.

Automation and Intelligent Control

- Control System Architecture
 - PLC Main Control Unit: Integrates real-time data collection, logic control, and fault diagnosis;
 - Human-Machine Interface: Touchscreen displays operational parameters (flow rate, energy consumption, water quality data), supporting manual/automatic mode switching;
 - Remote Monitoring: Data transmitted to a cloud platform via satellite/4G, enabling remote start/stop and alarm notifications (e.g., membrane clogging warnings).
- Intelligent Maintenance Strategy
 - Self-Cleaning Mechanism: Membrane system periodically backwashed, and bioreactor periodically air-water backwashed;
 - Fault Pre-Diagnosis: Vibration sensors monitor equipment anomalies, with AI algorithms predicting maintenance cycles (e.g., centrifuge bearing lubrication intervals).

Material and Durability Design

- Titanium Alloy Shell: Surface passivation treatment for seawater corrosion resistance (service life ≥ 20 years);
- Anti-Biofouling Coating: Fluoropolymer hydrophobic coating (contact angle $> 150^\circ$) to inhibit marine organism attachment;
- Modular Design: Quick disassembly and replacement of processing units to reduce maintenance costs.

Methods	Conceptual design solutions
FBStd	● Pollutant Removal: ○ Physical Filtration: Remove microplastics, sediments, and large debris. ○ Biological Treatment: Degrade organic waste (e.g., sewage, algae). ○ Chemical Treatment: Break down inorganic pollutants (e.g., heavy metals, oil spills). ○ Disinfection: Eliminate harmful bacteria (e.g., E. coli, Vibrio). ● Energy Efficiency: ○ Utilize renewable energy (solar, wave, or tidal power). ○ Minimize electricity consumption (e.g., passive flow systems). ● Eco-Friendly Operation: ○ No chemical additives that harm marine life. ○ Minimal sludge production (or biodegradable sludge). ● Cyclic Operation: ○ Continuous intake, treatment, and discharge without manual intervention. ○ Self-cleaning mechanisms to prevent biofouling. ● Modular & Scalable: ○ Can be deployed as floating units near coastal cities or attached to ships. ○ Adjustable capacity (e.g., 1,000–10,000 L/day). ● Low-Maintenance: ○ Automated backwashing for filters. ○ Corrosion-resistant materials (e.g., marine-grade stainless steel, HDPE). ● Hybrid Energy Supply: ○ Solar panels for daytime power. ○ Wave energy converters (WECs) for continuous operation. ● Closed-Loop System: ○ Treated water meets WHO/EPA standards before discharge. ○ Optional water reuse (e.g., for aquaculture or desalination).
KnowLoop	Structural Design ● Multi-layer Filtration System:

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- Membrane Filtration Layer: Utilize polyethersulfone membrane to intercept large-molecule pollutants.
 - Ultrafiltration Layer: Use ultrafiltration membrane to remove bacteria and viruses.
 - Nanofiltration Layer: Employ nanofiltration membrane to further eliminate organic matter and heavy metals.
 - Reverse Osmosis Layer: Apply reverse osmosis membrane to remove dissolved salts and fine particles.
 - Disinfection Module:
 - Chlorine Dioxide Disinfection: Integrate a chlorine dioxide generator for efficient pathogen elimination.
 - Disinfection Residue Testing: Equip with chemical analysis modules to ensure disinfectant residues meet safety standards.
 - Monitoring System:
 - Data Acquisition: Real-time collection of water quality parameters (e.g., pH, chemical oxygen demand, biochemical oxygen demand, coliform bacteria).
 - Microorganism Detection: Use flow cytometry to detect pathogens.
 - Sensor Technology: Integrate biosensors to monitor water quality anomalies.
 - Alarm System: Trigger alarms for water quality abnormalities, equipment failures, or leaks.
 - Safe Discharge Outlet:
 - Anti-backflow Design: Install check valves to prevent sewage backflow.
 - Anti-clogging Design: Equip with sedimentation tanks to prevent outlet clogging.
 - Corrosion-resistant Materials: Use plastics and composites to ensure corrosion resistance.
 - Outlet Cleaning: Regular automatic cleaning to prevent pollutant accumulation.
 - Sealed Treatment System:
 - Leak-proof Design: Use sealing rings, sealing gaskets, and sealing adhesives to ensure equipment sealing.
 - Corrosion-resistant Materials: Apply anti-corrosion coatings and fiberglass containers.
 - Pressure Control: Equip with pressure sensors and emergency discharge devices to prevent overpressure.
 - Gas Emission: Integrate emergency discharge devices to handle harmful gases generated during the process.
 - Sealing Inspection: Use ultrasonic testing for regular sealing checks.
 - Liquid Level Monitoring: Equip with liquid level alarms and emergency discharge devices to prevent overflow.

Functional Design

- Pathogen Removal:
 - Multi-layer Filtration: Gradually remove pathogens through membrane filtration, ultrafiltration, nanofiltration, and reverse osmosis.
 - Disinfection Treatment: Use chlorine dioxide disinfection to ensure complete pathogen inactivation.
 - Total Bacterial Count Testing: Apply flow cytometry to test the total bacterial count in treated water.
 - Leak Prevention:
 - Leak Detection: Conduct regular hydrostatic tests to ensure the equipment is leak-proof.
 - Overflow Protection: Equip with overflow protection devices to prevent sewage overflow.
 - Safe Discharge:
 - Pollutant Detection: Use biological detection technology to ensure discharged water is pollutant-free.
 - Discharge Report: Generate discharge reports to record the treatment process and discharge data.
 - Monitoring Equipment: Integrate data acquisition, water quality parameter monitoring, and microorganism detection modules.
 - Data Recording: Real-time recording of key data during the treatment process.
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- Abnormal Alarm: Trigger alarms for water quality anomalies, equipment failures, or leaks.
 - Environmental Compliance:
 - Industry Environmental Standards: Ensure the equipment complies with industry environmental standards.
 - Discharge Regulations: Meet national environmental regulations.

Operation and Maintenance

- User-friendly Operation: Equip with an intuitive interface to display water quality data and equipment status in real-time.
- Easy Maintenance: Modular design for easy replacement of filtration membranes and disinfection modules.
- Regular Inspections: Conduct regular sealing checks, pressure tests, and liquid level monitoring.

Materials and Manufacturing

- Material Selection: Use corrosion-resistant, high-temperature-resistant food-grade materials such as stainless steel, fiberglass, and plastic composites.
- Manufacturing Process: Apply precision manufacturing processes to ensure equipment sealing and durability.

Safety and Environmental Protection

- Safety Design: Equip with multiple safety protection measures, including pressure control, liquid level monitoring, and emergency discharge devices.
 - Environmental Design: Ensure treated water meets environmental standards, minimizing environmental impact.
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